torch-spyre Ecosystem Health Report

Date: 2026-05-27 | Pod: rganti-spyre-main | PyTorch: 2.11.0+cpu

3/3	FAIL	20	30
BUILD PASS	NIGHTLY CI	PRS MERGED (7D)	OPEN BUGS

What Needs Attention

1. **Nightly cat zero-tensor regression persists** — `test_cat_4d_dim2_zero` and `test_cat_4d_dim_m2_empty_first` failing for 5+ consecutive nightly runs (since at least 2026-05-23). PR #2259 ("Fix aten.cat lowering crash with zero-sized tensor inputs", merged 2026-05-22) does NOT cover these cases. No open issue or fix PR. Needs ownership.
2. **PR #2218 (Upgrade to PT 2.12) idle** — Open since 2026-05-21 by ani300, REVIEW_REQUIRED, no recent reviewer activity. Blocks future PT alignment.
3. **23 PRs stalled >7 days** — Substantial review backlog, including #2107 (LayerNorm fp32 fix), #2059 (CI gate-job suite reporting), #2128 (xfail xdist SIGSEGV), #2055 (xfail fp32 lx-planning).
4. **26 open model-enablement bugs across Granite/Mistral/Llama/Ministral** — Mostly INT32/INT64 dtype and fp32 op gaps. Driving June2026 milestone. No new bugs in last 24h.

Build Status

Component	Commit	Status	Notes
deeptools	`940d8d34fe` — [DDC/ddl] subchunking of across-stick reduction for sum/mean/...	PASS	cmake + make -j16 install (~10 min)
flex	`a233ca5` — Add region_id to segment_id mapping to FlexAllocator (#1034)	PASS	cmake + make -j16 install (~1 min)
torch-spyre	`f182ab3` — add JobPlan launch (#2013)	PASS	uv sync editable install (~2 min)

CI Status

Workflow	Latest	Status	Trend
Spyre Hardware Tests	2026-05-27	PASS	Healthy (3/3 recent green)
Linters	2026-05-27	PASS	Healthy
Upstream PyTorch Tests	2026-05-27	PASS	Healthy
Nightly Tests	2026-05-27	FAIL	**Persistently failing 5+ consecutive days (cat zero-tensor)**

Nightly Failure Triage

Failure 1 & 2: aten.cat with zero-sized tensors — PERSISTENT REGRESSION

Tests: `test_cat_4d_dim2_zero`, `test_cat_4d_dim_m2_empty_first`

Error: `InductorError: LoweringException: AssertionError` in `torch/_inductor/lowering.py:2177`

Repro signature: args[0]=[`FixedLayout('cpu', float16, size=[0], stride=[1])`, `FixedLayout('cpu', float16, size=[1, 8, 14, 64], stride=[7168, 896, 64, 1])`], args[1]=2 (or -2)

Root cause: Empty 1D tensor passed in alongside a 4D tensor; cat along dim=2/-2 fails because the empty tensor only has dim 0. PR #2259 (merged 2026-05-22) addressed cat zero-sized crash but not this dim-mismatch path. Fix needs to either filter zero-numel tensors before lowering or reshape the empty tensor to match.

Duration: Failing 2026-05-23, 24, 25, 26, 27 (5 consecutive nightlies confirmed)

Assessment: **Regression — no open issue, no fix PR, no owner**

Overall nightly: 2 failed, 2293 passed, 54 skipped, 1048 deselected, 321 xfailed, 2 xpassed. Pass rate: 99.91%. Note: yesterday's nightly also had 1 flaky `test_layernorm_2d_lx_planning_reduction` failure but that did NOT recur today.

Version Alignment

Check	Expected	Actual	Status
PyTorch	`torch~=2.11.0` (pyproject.toml)	`2.11.0+cpu`	MATCH
transformers (post-build)	`>=5.0` (hf_adapters)	`4.57.6` after uv sync; bumped to `5.9.0` for E2E	DRIFT (POST-INSTALL)

Repo Drift (last 7 days)

Repo	Commits	API-touching	Risk
deeptools	81	15+ (DCC, BaseOpt, RcuOpt, DataOp, DSM internals)	Low — internal compiler changes, no torch-spyre API surface affected (build clean)
flex	12	8 (FlexAllocator MemoryType/region_id, profiler API consolidation)	Medium — FlexAllocator API additions could affect future torch-spyre work; build green today
torch-spyre	62	—	—

Blocking PRs / Notable Open

PR	Title	Why Notable	Status
#2218	Upgrade to PT 2.12	Future toolchain alignment	REVIEW_REQUIRED (6d open)
#2107	Fix layernormscale output scale for fp32 LayerNorm	Fixes #1980 LayerNorm correctness bug	REVIEW_REQUIRED (12d open, stalled)
#2298	Fix core-div-mismatch logic with PerCoreView	Backend correctness (DRAFT)	DRAFT (1d)
#2261	Clean up spyre::overwrite usage	Touches cat path; potential nightly fix candidate	REVIEW_REQUIRED
#2250	Coarse-Tiling Loop IR for the Spyre Backend	Major IR change, dgrove-oss	REVIEW_REQUIRED (5d)
#2003	Add utility to support PyTorch mark_dynamic API with max	Dynamic shapes support	REVIEW_REQUIRED (16d, stalled)

Project Velocity (last 7 days)

Metric	Count
PRs merged	20
Open PRs (non-draft)	~35
Stalled PRs (no activity >7d)	23

Recent merges include	#2319, #2318, #2312, #2307 (docs drift fix), #2303, #2301, #2297, #2283, #2277, #2274, #2273, #2270, #2269, #2268, #2267, #2263, #2259 (cat zero-size), #2251, #2299 (Inductor parallel shards)

Open Bugs

Metric	Value
Total open bugs	30
New (last 24h)	0
Closed in last 7 days	7+ (e.g. #2087, #2070, #1968, #1886, #1884, #1881, #1850)
Critical/priority labels	None set (no priority labels in use)

Bug Distribution (top categories)

Category	Count	Examples
Model enablement (Granite/Mistral/Llama/Ministral)	~18	#1849 add+alpha, #1880/#1942 index_copy_, #1971 pow RMSNorm, #1970 linear, #1969 rsqrt, #1837 max, #1497/#1482 to/float, #2207 Ministral stick expr
Inductor / propagate_hints	4	#2247 hint loss after AOT mutations, #2246 thread-safety, #2166 softmax lx perf, #2195 MHA compile
Backend compiler	1	#1996 dl16tofp32 zeroed elements wider than 16
Numerical correctness	3	#1980 LayerNorm elementwise_affine=False, #1919 contiguous, #1918 matmul large shapes
Op coverage gaps	4	#1485/#1488/#1489 all/eq, #1714 unsqueeze int64

Smoke Test (post-build)

Stage	Result	Detail
Import torch_spyre	PASS	Spyre available=True, device_count=1
Eager add (fp16, 4×4)	PASS	shape=(4,4)
Compiled add (fp16, 128×128)	PASS	torch.compile() default Inductor backend on Spyre
E2E Qwen3 0.6B (5 tokens)	PASS	Load: 6.1s, Generate: 79.8s, Output: " Paris. The capital of"

Cross-Repo API Mismatches

None detected. All three components built clean against their respective HEADs (deeptools master, flex main, torch-spyre main).
